Accent Classifier

CS663 Final Presentation

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12 May 2026

Purpose: classify a speaker's accent

Unique Elements:

- Variational Autoencoder (VAE)
- Multi-model comparison

EDA - GLOBEv3

Built on the Mozilla Data Collective (MDC)'s Common Voice 21.0 English (CV) Dataset

Includes preprocessing

Improves on issues with CV features

GLOBEv3

A Step Forward In Quality

Very Large Dataset

~349GB with audio

~723k rows

251 unique accents

Common Voice Improvements

Higher Audio Quality

- Filtered low SNR

Greater transcriptions

- Whisper used, lower WER

To Compare:

One accent in CV was "not bad".

Yes.

The accent was labeled as "not bad"

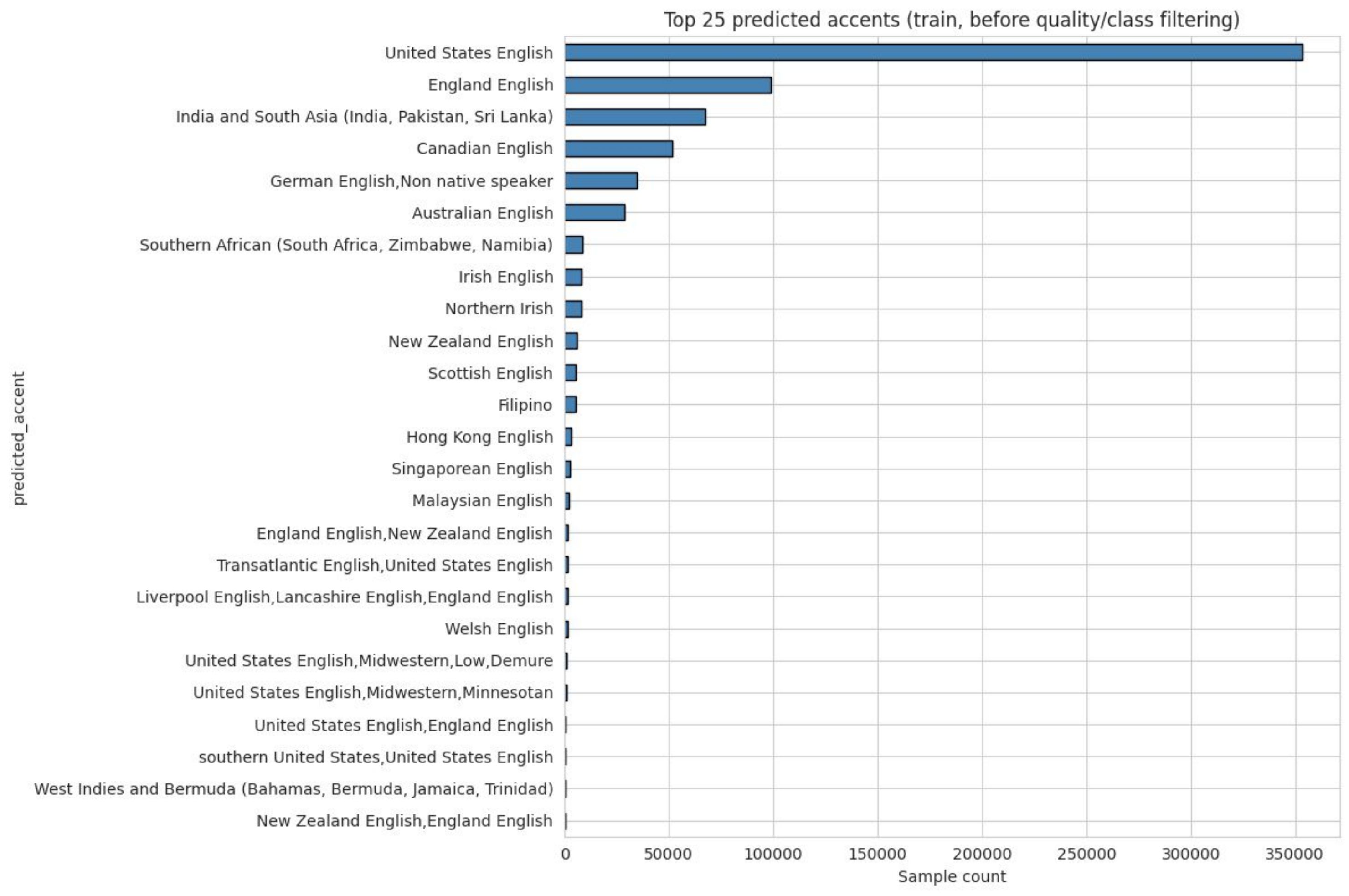
Still Has Issues

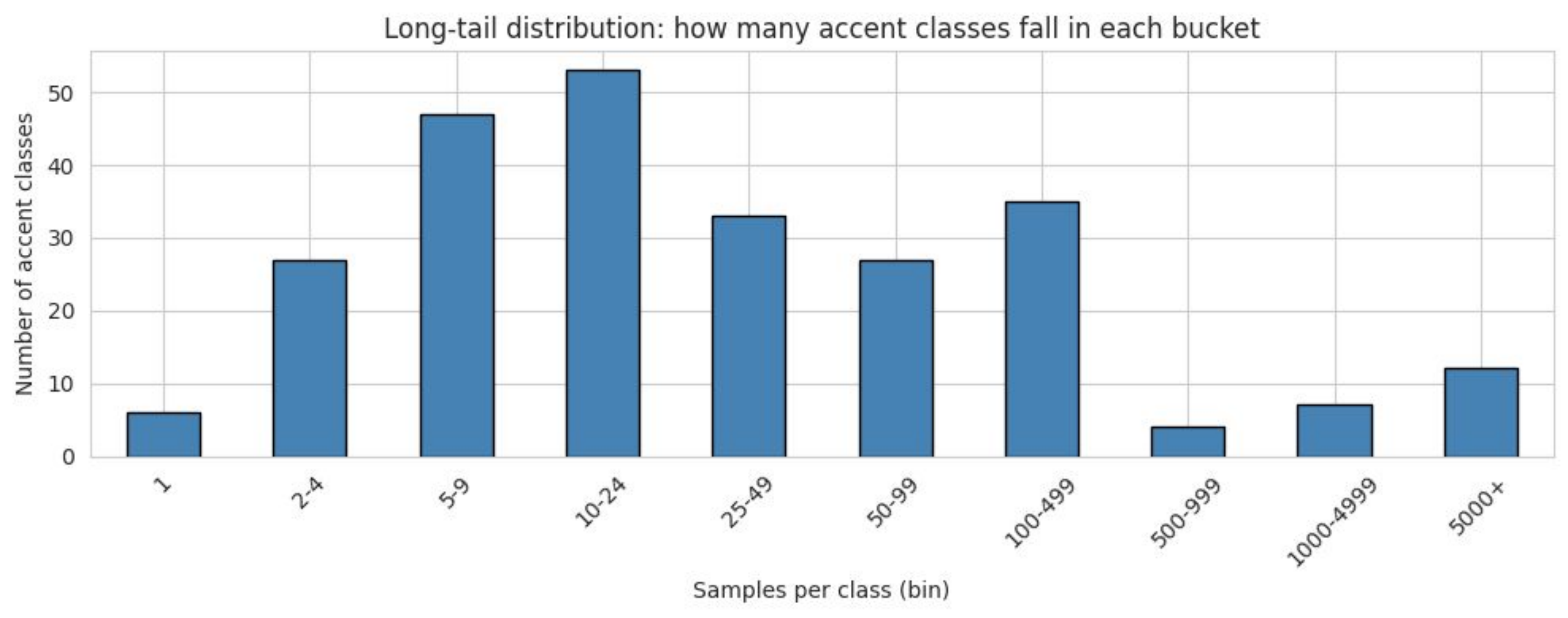
58 classes

99.4% of the data (classes with ≥ 100)

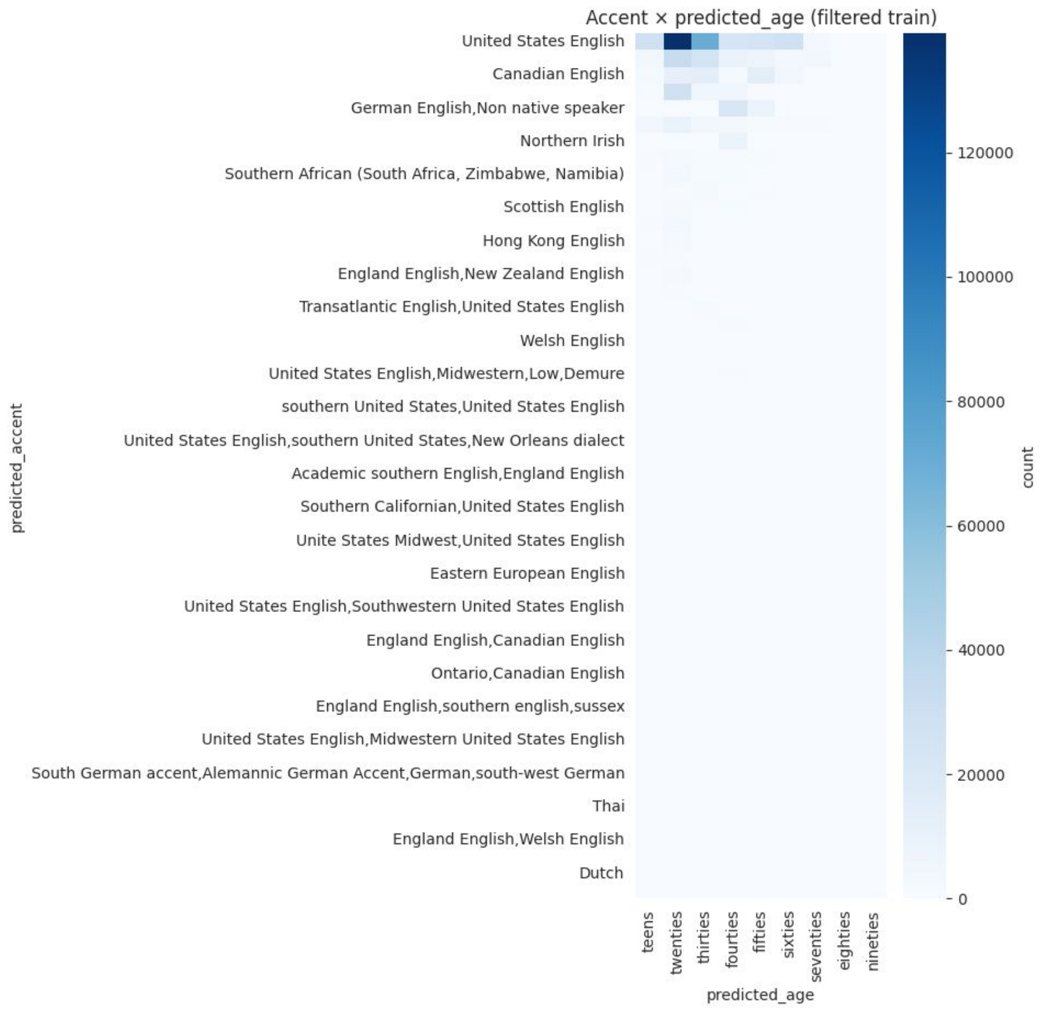
Highly right skewed to younger age groups

Some accents fall under one age group/gender completely


Top 25 predicted accents (train, before quality/class filtering)
United States English
England English
India and South Asia (India, Pakistan, Sri Lanka)
Canadian English
German English,Non native speaker
Australian English
Southern African (South Africa, Zimbabwe, Namibia)
Irish English
Northern Irish
New Zealand English
Scottish English
Filipino
Hong Kong English
Singaporean English
Malaysian English
England English,New Zealand English
Transatlantic English,United States English
Liverpool English,Lancashire English,England English
Welsh English
United States English,Midwestern,Low,Demure
United States English,Midwestern,Minnesotan
United States English,England English
southern United States,United States English
West Indies and Bermuda (Bahamas, Bermuda, Jamaica, Trinidad)
New Zealand English,England English
predicted_accent
0
50000
100000
150000
200000
250000
300000
350000
Sample count


Long-tail distribution: how many accent classes fall in each bucket
Number of accent classes
0
10
20
30
40
50
1
2-4
5-9
10-24
25-49
50-99
100-499
500-999
1000-4999
5000+
Samples per class (bin)


Accent × predicted_gender (filtered train)
United States English
Canadian English
German English,Non native speaker
Northern Irish
Southern African (South Africa, Zimbabwe, Namibia)
Scottish English
Hong Kong English
England English,New Zealand English
Transatlantic English,United States English
Welsh English
United States English,Midwestern,Low,Demure
southern United States,United States English
United States English,southern United States,New Orleans dialect
Academic southern English,England English
Southern Californian,United States English
Unite States Midwest,United States English
Eastern European English
United States English,Southwestern United States English
England English,Canadian English
Ontario,Canadian English
England English,southern english,sussex
United States English,Midwestern United States English
South German accent,Alemannic German Accent,German,south-west German
Thai
England English,Welsh English
Dutch
predicted_accent
female
male
predicted_gender
count
200000
175000
150000
125000
100000
75000
50000
25000
0



Accent × predicted_age (filtered train)
United States English
Canadian English
German English,Non native speaker
Northern Irish
Southern African (South Africa, Zimbabwe, Namibia)
Scottish English
Hong Kong English
England English,New Zealand English
Transatlantic English,United States English
Welsh English
United States English,Midwestern,Low,Demure
southern United States,United States English
United States English,southern United States,New Orleans dialect
Academic southern English,England English
Southern Californian,United States English
Unite States Midwest,United States English
Eastern European English
United States English,Southwestern United States English
England English,Canadian English
Ontario,Canadian English
England English,southern english,sussex
United States English,Midwestern United States English
South German accent,Alemannic German Accent,German,south-west German
Thai
England English,Welsh English
Dutch
predicted_accent
teens
twenties
thirties
fourties
fifties
sixties
seventies
eighties
nineties
predicted_age
120000
100000
80000
60000
40000
20000
0
count

Related Works

Discuss about papers we looked into and some of the shared challenges that we faced that the papers also faced.

Related Work

Accented Speech Recognition: A Survey

- 2021 survey by Hinsvark et al.
- ASR systems perform well on common accents but struggle with underrepresented ones
- No standard benchmark exists, which makes it hard to compare models
- Class imbalance is a known, persistent challenge

Exploring Accent Similarity for Cross-Accented Speech Recognition

- 2024 paper by Gu et al.
- Proposes using accent similarity to transfer knowledge to low-resource accents
- Models can “borrow” from similar, data-rich accents to improve on rare ones
- Confirms that low-resource accent performance is a hard, unresolved problem

Related Work

A Systematic Review of Accent Classification Techniques

- 2026 systematic review by Salifu et al.
- Reviewed 58 studies and 24 datasets on accent classification

Speech Accent Archive

- The most cited dataset in the field
- Used in 36 out of 58 papers

Feature Extraction

- MFCC – a feature extraction technique in speech & audio signal processing esp in applications (e.g. speech and emotion recognition and speech verification)
  - We went a different route: mel spectograms

CNNs (Convolutional Neural Networks)

- CNNs are the dominant models
- Newer work uses Wav2Vec and Transformers

Speech Accent Archive

Dataset Structure

- From George Mason University.
- Contains 2,140 speech samples, speakers from 177 different countries and have 214 native languages.
- All speaking the same passage in English.
- Mp3 audio file format.
- Contains demographic information for each speaker.

https://www.kaggle.com/datasets/rtatman/speech-accent-archive

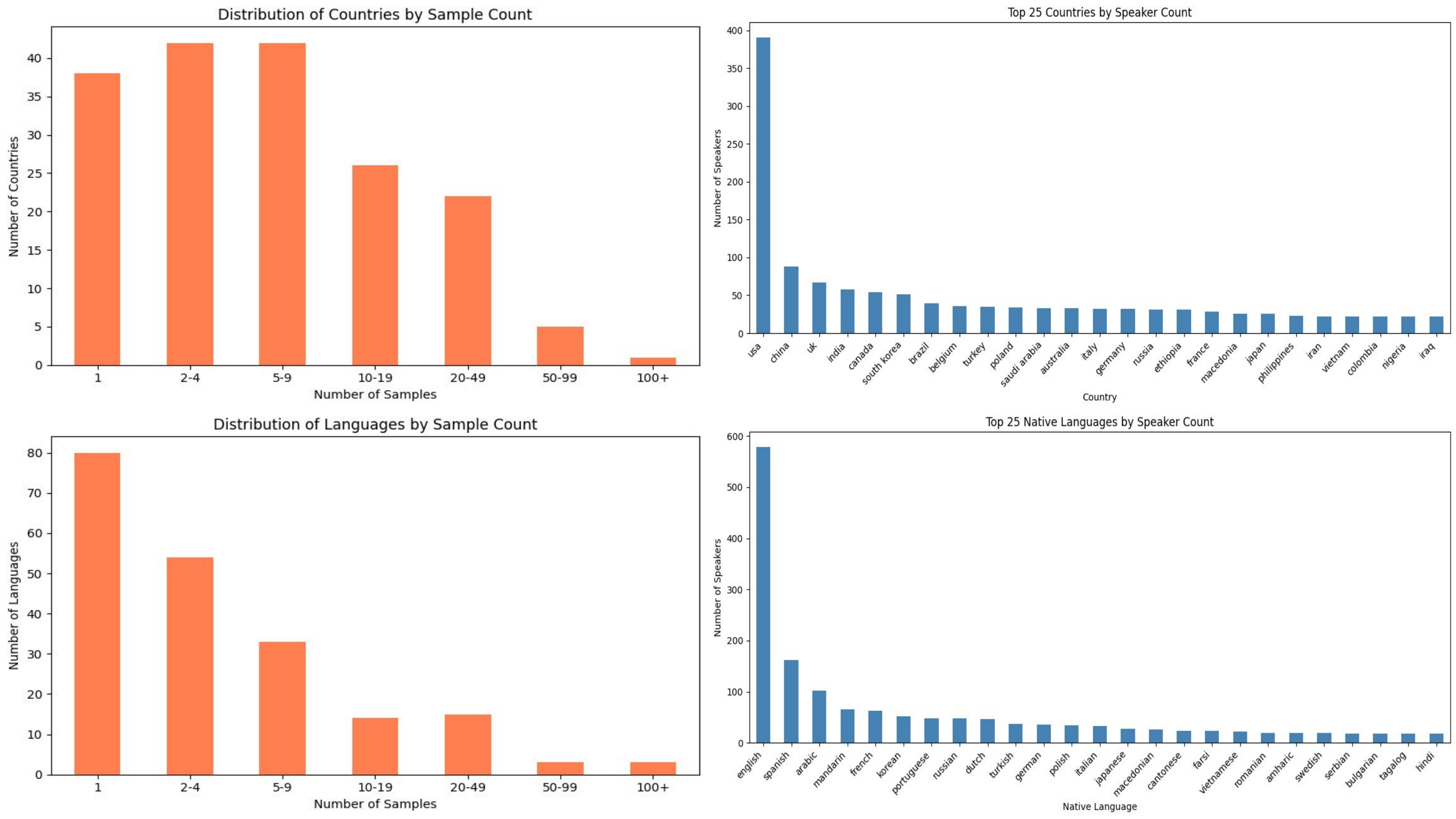
Dataset Exploration

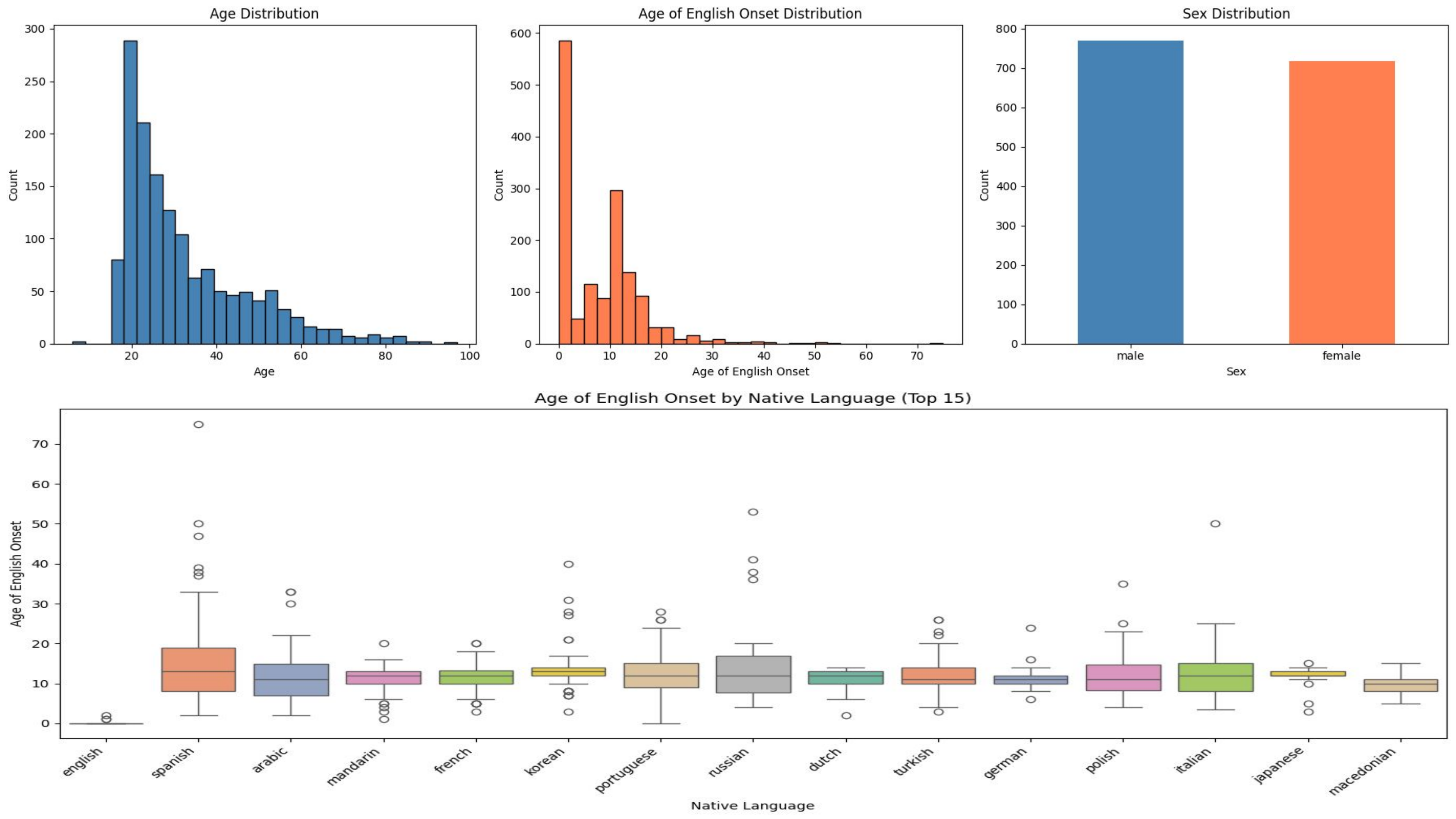
Cleaning

- 2,172 rows with 12 columns
- File_missing column signifies a missing corresponding audio file for that speaker
- 3 unnamed columns filled with NaN
- 32 missing the audio file and 9 missing various demographic information
- No features engineered.

Analysis

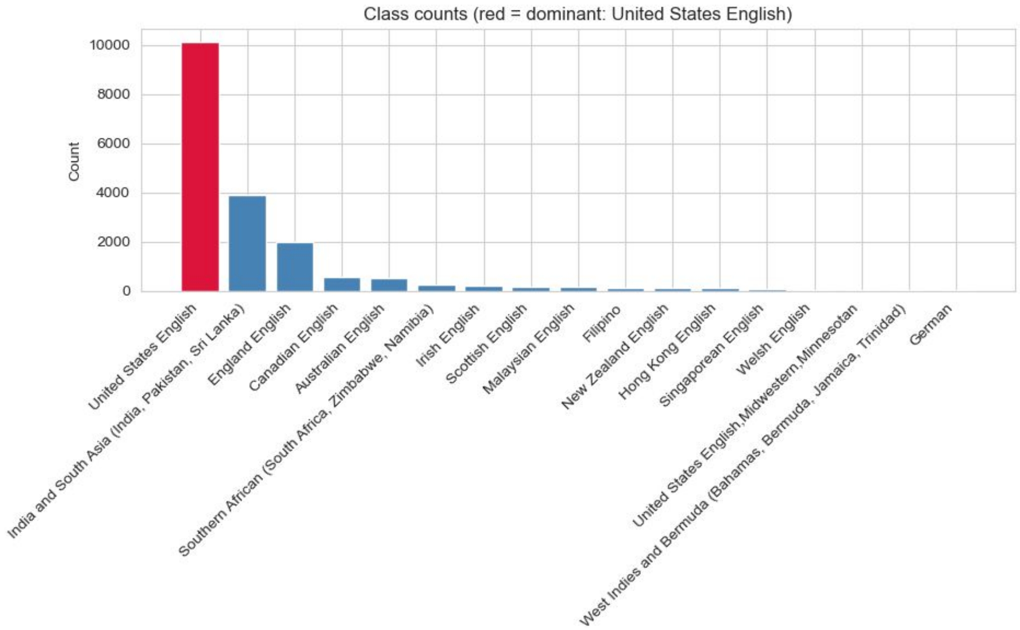
- Male-Female split is fairly even, 1103-1037.
- Typical age between 20-35.
- English native language and USA nationality dominates the dataset.
- Significant number of countries/languages have less than 20 samples.
- Problem: Large class imbalance for countries and native languages.
- Solution: **Focus on countries and languages with at least 20 samples.**
- Overall quality is fine to use with precautions for the noted limitations.

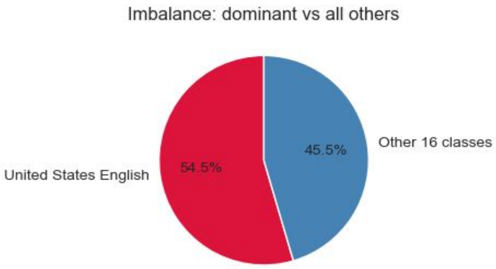

Distribution of Countries by Sample Count
Number of Countries
Number of Samples
1
2-4
5-9
10-19
20-49
50-99
100+
Top 25 Countries by Speaker Count
Number of Speakers
Country
usa
china
uk
india
canada
south korea
brazil
belgium
turkey
poland
saudi arabia
australia
italy
germany
russia
ethiopia
france
macedonia
japan
philippines
iran
vietnam
colombia
nigeria
iraq
Distribution of Languages by Sample Count
Number of Languages
Number of Samples
Top 25 Native Languages by Speaker Count
Number of Speakers
Native Language
english
spanish
arabic
mandarin
french
korean
portuguese
russian
dutch
turkish
german
polish
italian
japanese
macedonian
cantonese
farsi
vietnamese
romanian
amharic
swedish
serbian
bulgarian
tagalog
hindi



Age Distribution
Count
Age
Age of English Onset Distribution
Count
Age of English Onset
Sex Distribution
Count
male
female
Sex
Age of English Onset by Native Language (Top 15)
Age of English Onset
english
spanish
arabic
mandarin
french
korean
portuguese
russian
dutch
turkish
german
polish
italian
japanese
macedonian
Native Language

Class imbalance





Why a VAE gate?

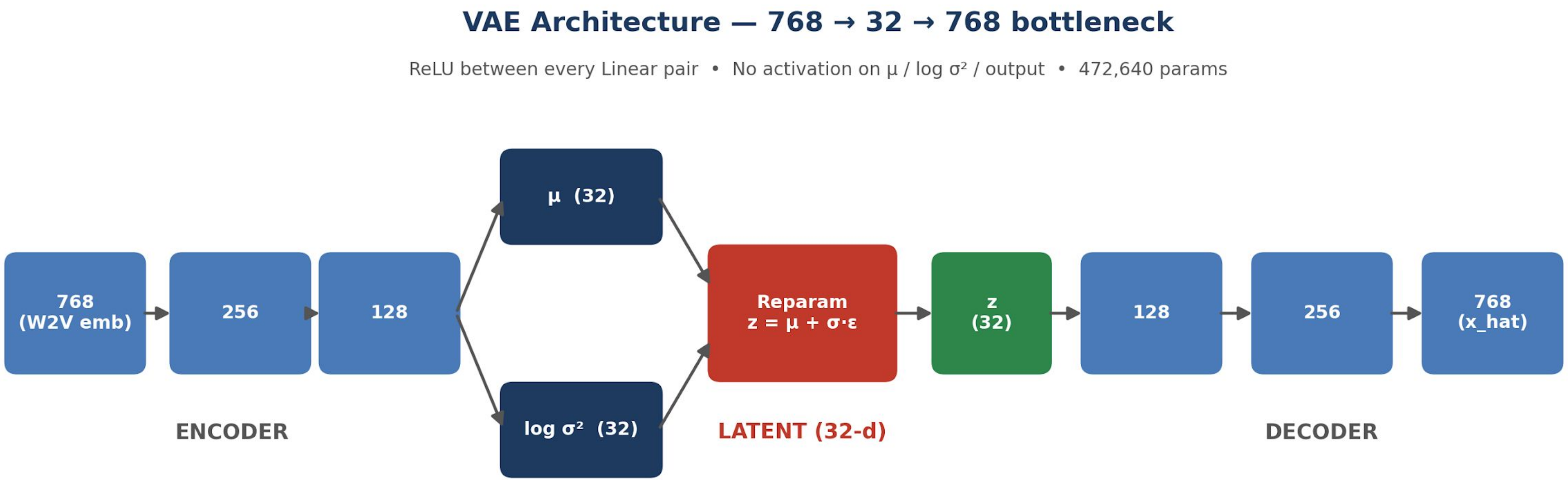
17 accent classes; **U.S. English = 54.5%** of all data

End-to-end classifiers collapse to predicting USE

Fix: train a VAE *only* on USE → use recon error as a gate

- `recon error < τ` → predict USE
- `recon error ≥ τ` → forward to 16-class classifier

VAE architecture



Reparameterization Trick

```
def reparam(self, mu, logvar):
    std = (0.5 * logvar).exp()            # σ = exp(½ · log σ²)
    eps = torch.randn_like(std)            # ε ~ N(0, I)
    return mu + std * eps                 # z = μ + σ · ε


def forward(self, x):
    mu, logvar = self.encode(x)            # encoder → distribution params
    std = (0.5 * logvar).exp()            # σ = exp(½ · log σ²)
    eps = torch.randn_like(std)            # ε ~ N(0, I)
    z = mu + std * eps                   # reparameterization trick
    x_hat = self.decode(z)                # decoder reconstructs
    return x_hat, mu, logvar
```

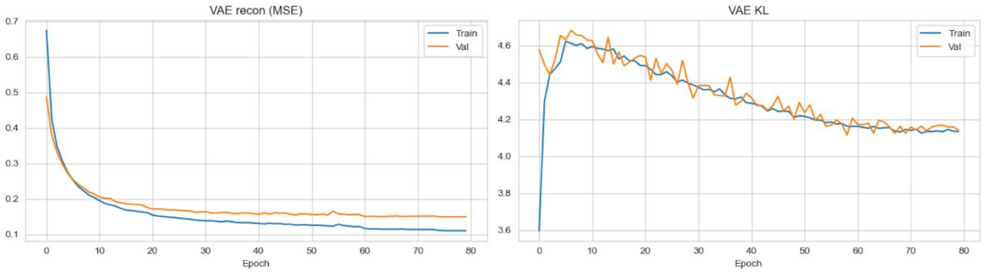
Loss function — β-VAE objective

$$\mathcal{L} = \mathrm{MSE}(\hat{x}, x) + \beta \cdot \mathrm{KL}(\, q(z \mid x) \parallel \mathcal{N}(0, I)\,)$$

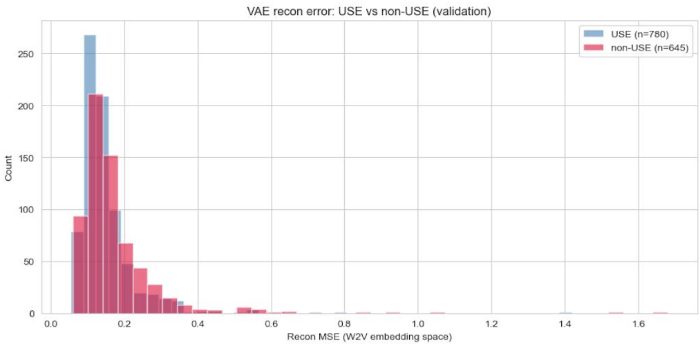
reconstruction term + weighted KL regularizer

Training + reconstruction error

VAE training (loss + KL over 80 epochs)



Recon error: USE vs non-USE (val)



Threshold τ and gate performance

- **τ = 0.124** (balanced-acc-optimal)
- **ROC-AUC = 0.593** (USE vs non-USE)
- Test USE-recall **0.427** · non-USE-recall **0.705**
- Gate works — better than chance — but below AUC ≥ 0.70 target
- USE-recall caps every downstream classifier at 0.427

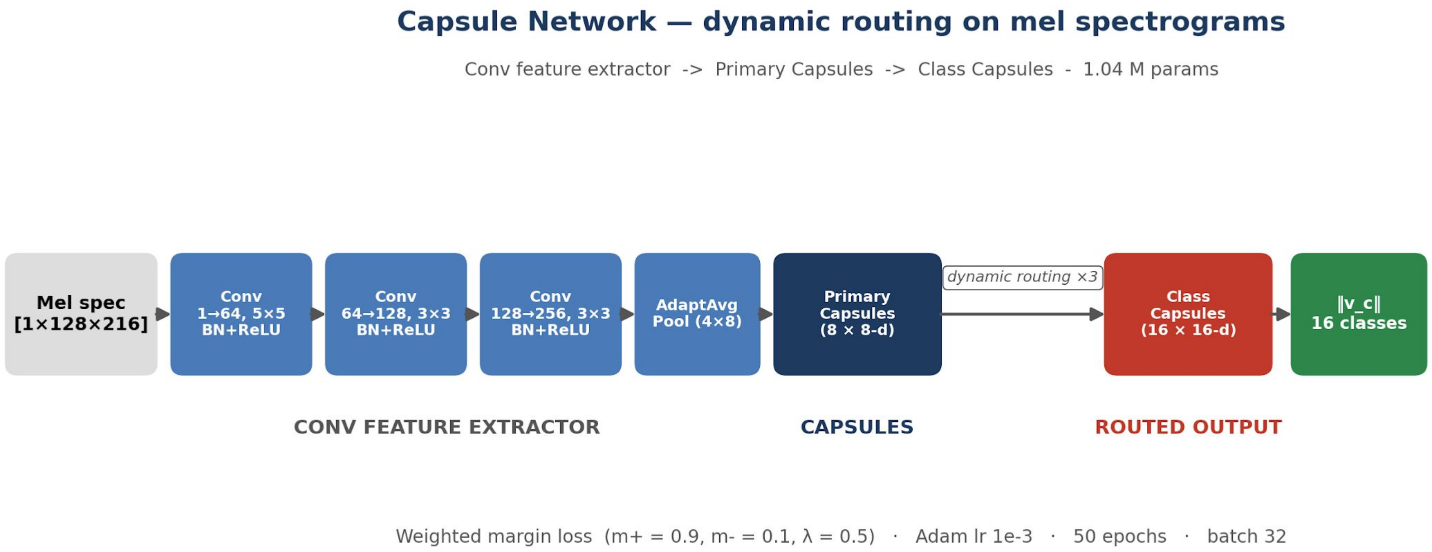
Downstream classifiers — 4 architectures compared

Once the gate routes a sample to "non-USE," it goes to one of four classifiers trained on the 16-class non-USE subset:

- **Capsule Network** — mel-spec CNN + dynamic routing
- **EfficientNet-B0** — ImageNet transfer learning on mel-spec
- **Wav2Vec 2.0 head** — frozen W2V features + MLP
- **AST head** — frozen Audio Spectrogram Transformer + MLP

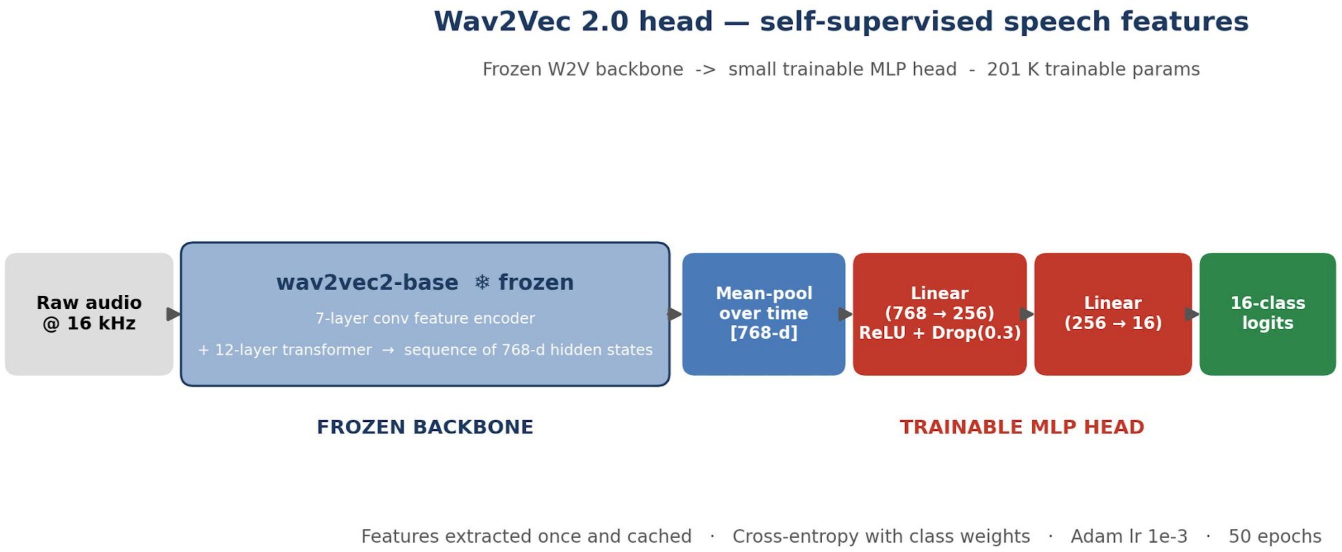
All four use the same class-weighted training setup for a fair comparison.

Classifier 1 — Capsule Network

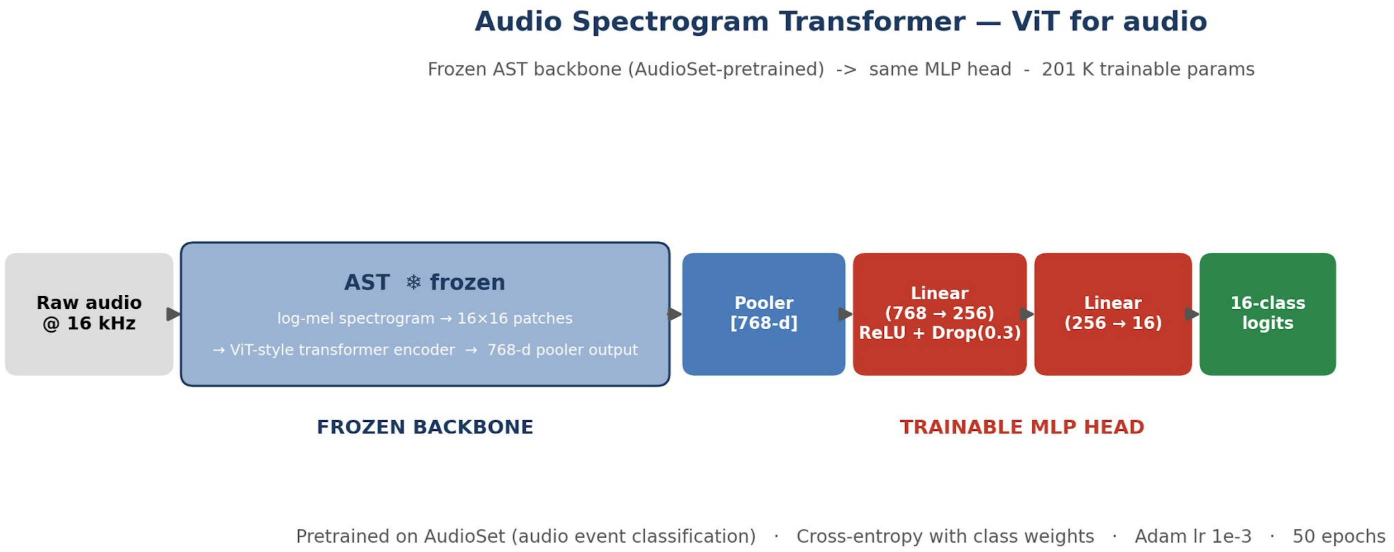


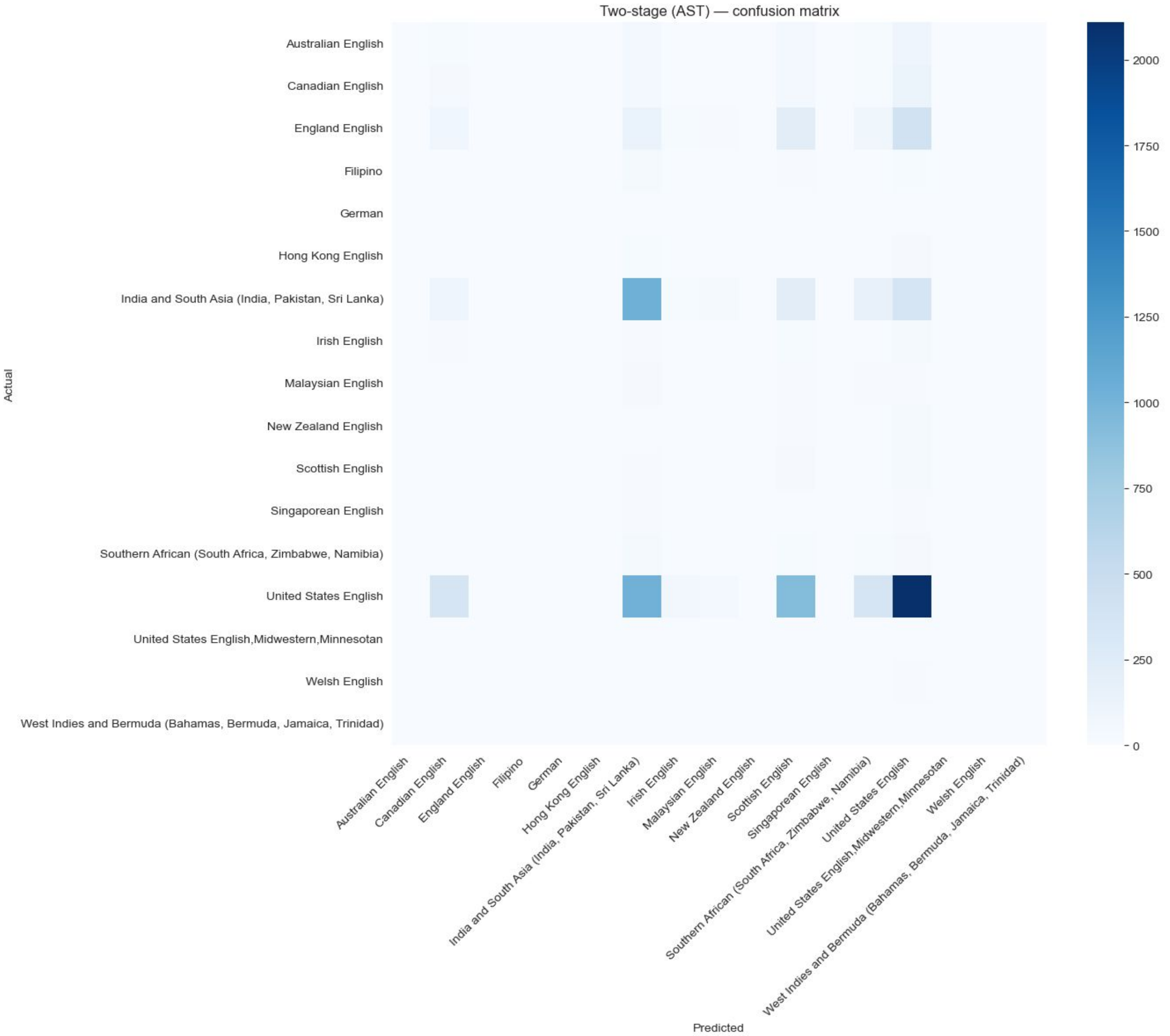
Classifier 2 — EfficientNet-B0

Classifier 3 — Wav2Vec 2.0 head

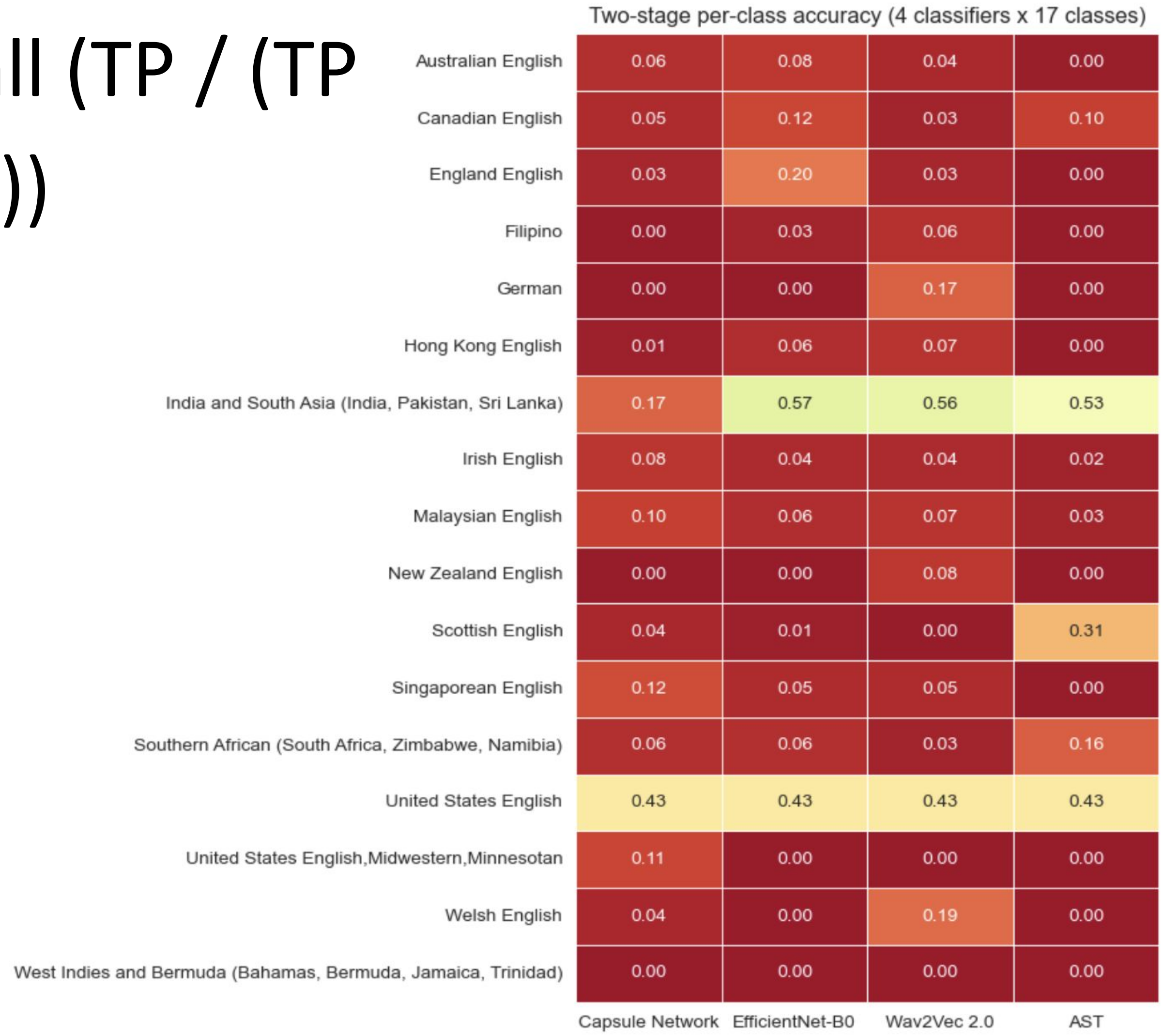


Classifier 4 — Audio Spectrogram Transformer




Two-stage (AST) — confusion matrix
Actual
Predicted
Australian English
Canadian English
England English
Filipino
German
Hong Kong English
India and South Asia (India, Pakistan, Sri Lanka)
Irish English
Malaysian English
New Zealand English
Scottish English
Singaporean English
Southern African (South Africa, Zimbabwe, Namibia)
United States English
United States English,Midwestern,Minnesotan
Welsh English
West Indies and Bermuda (Bahamas, Bermuda, Jamaica, Trinidad)
0
250
500
750
1000
1250
1500
1750
2000

Recall (TP / (TP + FN))



Future work

- Replace generative gate with a **supervised binary head** on the same W2V embeddings → expected AUC > 0.85
- Sweep $\beta \in \{1e\text{-}4, 1e\text{-}2\}$ and latent_dim $\in \{64, 128\}$ to test if weakness is capacity or signal-to-noise
- Augment the long-tail accents (SpecAugment, pitch shift)